

ELECTRIC CHARGES AND FIELDS

“ Charge is the source, Field is the effect. ”

1.1 Electric Charge

* Electric charge is a **fundamental property** of matter that causes it to experience a force when placed in an **electric field**.

* SI Unit : **Coulomb (C)**

* Nature : Two types → **Positive (+)** and **Negative (-)**

* Charge is always **conserved**.

* Key Points

- Like charges repel
- Unlike charges attract
- Charge quantization :

$$q = \pm ne$$

$n = \text{integer}$

$$e = 1.6 \times 10^{-19} \text{ C}$$

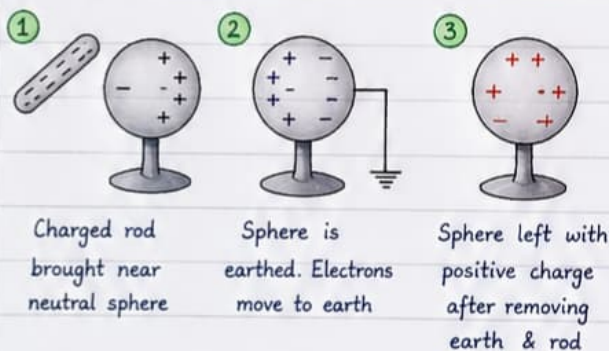
1.2 Methods of Charging

① **By Friction** → When two bodies are rubbed, charge is transferred.

② **By Conduction** → When a charged body is brought in contact with a neutral body.

③ **By Induction** → When a charged body influences a nearby neutral body without touching it.

* Charging by Induction (Example)



1.3 Coulomb's Law

* The electrostatic force between two point charges is **directly proportional** to the product of charges and **inversely proportional** to the square of the distance between them.

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}$$

where,

F = electrostatic force (N)

q_1, q_2 = charges (C)

r = distance between charges (m)

ϵ_0 = permittivity of free space

$$= 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

* Direction of Force

→ If q_1 and q_2 are of same sign → **Repulsion**



→ If q_1 and q_2 are of opposite sign → **Attraction**



1.4 Charge is Quantized

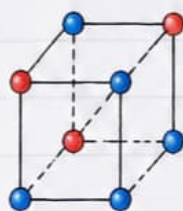
* The smallest unit of charge is $e = 1.6 \times 10^{-19} \text{ C}$

* Any charge $q = \pm ne$, where $n = 0, 1, 2, 3 \dots$

This means charge exists in **discrete packets**.

Quick Revision

- * Charge → Fundamental property (+, -) ✓
- * Charge is conserved ✓
- * Coulomb's Law → $F \propto q_1 q_2$ and $F \propto \frac{1}{r^2}$ ✓
- * Charge is quantized → $q = \pm ne$ ✓



Symmetric arrangement of charges

Real Life Example

Lightning is a natural example of electric charge. Clouds get charged, and a huge charge transfer takes place between cloud and ground.



1.5 Electric Field

- * The electric field at a point is the electric force experienced by a unit positive charge placed at that point.
- * It is a **vector** quantity.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

where,

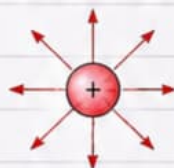
$\vec{E}$ = electric field (NC^{-1} or Vm^{-1})

$\vec{F}$ = electric force (N)

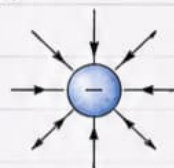
q_0 = unit positive charge (C)

♦ Electric Field Lines

- * Imaginary lines used to represent electric field.
- * The tangent at any point gives the direction of $\vec{E}$ at that point.
- * Originates from + charge and terminates on - charge.
- * No two field lines intersect each other.
- * The density of lines shows the strength of field (closer $\rightarrow$ stronger).



Field lines due to + charge



Field lines due to - charge

1.6 Electric Field due to a Point Charge

- * The electric field at a distance r from a point charge q is given by Coulomb's law.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

where,

$\hat{r}$ = unit vector from charge to the point (radially outward for +q, inward for -q)

q = source charge (C)

r = distance from charge (m)

ϵ_0 = permittivity of free space

$$= 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

For Positive Charge (+q)



Field is directed radially outward from the charge.

For Negative Charge (-q)

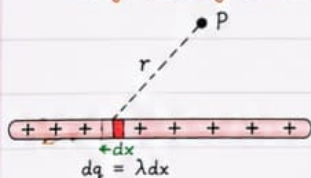


Field is directed radially inward towards the charge.

1.7 Electric Field due to Continuous Charge Distributions

- * Electric field at a point due to many charges is given by superposition principle.

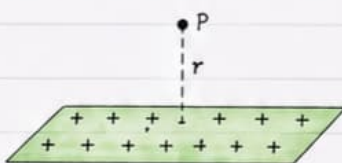
- (ii) Due to a Charged Line (Linear Charge Density $\lambda \text{ C m}^{-1}$)



$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

(for an infinitely long line)

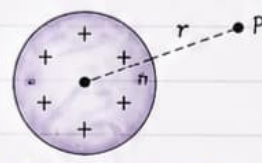
- (iii) Due to a Charged Surface (Surface Charge Density $\sigma \text{ C m}^{-2}$)



$$E = \frac{\sigma}{2\epsilon_0}$$

(for an infinitely large sheet)

- (iii) Due to a Charged Volume (Volume Charge Density $\rho \text{ C m}^{-3}$)



$$E = \frac{\rho r}{3\epsilon_0}$$

(inside a uniformly charged sphere)

Quick Points

- * Electric field is vector $\rightarrow$ has magnitude & direction.
- * SI unit $\rightarrow \text{NC}^{-1}$ or Vm^{-1} .
- * Field lines help in visualizing the field.

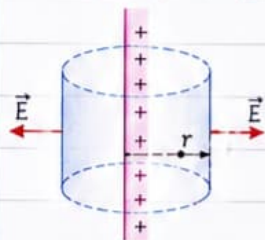
- * For + charge $\rightarrow$ lines go out.
- * For - charge $\rightarrow$ lines come in.
- * Field lines never form closed loops.

Stronger field $\rightarrow$ more lines closer together.

Remember $\rightarrow$ Charge creates the field and field exerts force on charge. ★

1.8 Electric Field due to Some Common Charge Distributions

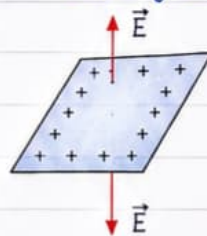
(i) Due to an infinitely long straight uniformly charged line ($\lambda \text{ C m}^{-1}$)



$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Directed radially outward
(for $+\lambda$)

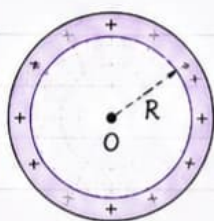
(ii) Due to an infinitely large uniformly charged plane sheet ($\sigma \text{ C m}^{-2}$)



$$E = \frac{\sigma}{2\epsilon_0}$$

Directed perpendicular
to the sheet
(away from $+\sigma$)

(iii) Due to a uniformly charged spherical shell (surface charge density σ)

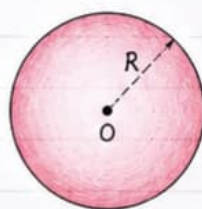


$$E = 0 \text{ for } r < R$$

$$E = \frac{\sigma R^2}{\epsilon_0 r^2} \text{ for } r > R$$

(acts as if charge is at centre
for $r > R$)

(iv) Due to a uniformly charged solid sphere (volume charge density ρ)



$$E = \frac{\rho r}{3\epsilon_0} \text{ for } r \leq R$$

$$E = \frac{\rho R^3}{3\epsilon_0 r^2} \text{ for } r \geq R$$

Field increases linearly inside
and behaves as point charge
outside.

1.9 Electric Flux

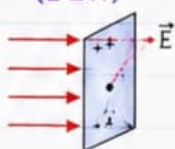
- * The electric flux through a surface is the measure of the number of electric field lines passing through that surface.
- * For a small area vector $d\vec{A}$ making an angle θ with electric field $\vec{E}$

$$d\phi_E = \vec{E} \cdot d\vec{A} = E dA \cos \theta$$

SI Unit of flux = $\text{N m}^2 \text{C}^{-1}$

Flux through a Plane Surface (Area A)

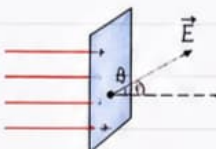
(i) $\theta = 0^\circ$
($\vec{E} \perp \vec{A}$)



$$\phi_E = EA$$

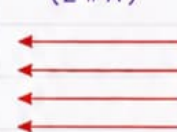
(Maximum)

(ii) $0^\circ < \theta < 90^\circ$



$$\phi_E = EA \cos \theta$$

(iii) $\theta = 90^\circ$
($\vec{E} \parallel \vec{A}$)



$$\phi_E = 0$$

(Minimum)

1.10 Gauss's Law

- * The total electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ times the net charge enclosed by the surface.

Mathematical form

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

where,

$\oint \vec{E} \cdot d\vec{A}$ = total electric flux through closed surface

q_{enclosed} = net charge enclosed

ϵ_0 = permittivity of free space

Quick Revision

- * Electric field is a vector quantity. ✓
- * Field lines start from + and end on - charge. ✓
- * Closer the lines, stronger the field. ✓
- * Electric flux depends on angle between $\vec{E}$ and $\vec{A}$. ✓
- * Gauss's law is valid for any closed surface. ✓

Applications of Gauss's Law

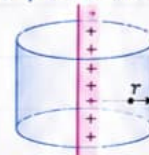
(i) To find E due to a point charge



$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

(radially outward for $+q$)

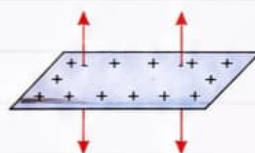
(ii) To find E due to an infinite line charge (λ)



$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

(radially outward)

(iii) To find E due to an infinite plane sheet (σ)



$$E = \frac{\sigma}{2\epsilon_0}$$

(perpendicular to sheet)

Remember → Field describes the effect charge produces, while force is the result of that effect on another charge. ★

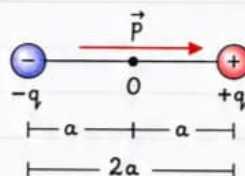
1.11 Electric Dipole

- * An electric dipole consists of two equal and opposite charges (+q and -q) separated by a small distance 2a.

- * Dipole moment $\vec{p}$ is defined as

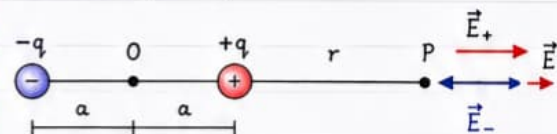
$$\vec{p} = q \cdot 2\vec{a}$$

Direction of $\vec{p}$ is from -q to +q.



Field due to an Electric Dipole (on Axial Line)

At a point P on the axial line at a distance r from the centre O ($r \gg a$)

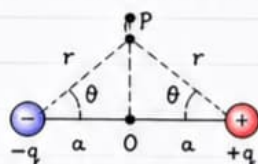


Net electric field at P,

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \quad (\text{along } \vec{p})$$

Field due to an Electric Dipole (on Equatorial Line)

At a point P on the equatorial line at distance r from O ($r \gg a$)



Net electric field at P,

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \quad (\text{opposite to } \vec{p})$$

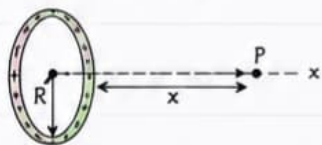
1.12 Continuous Charge Distributions

When charge is distributed continuously, we use charge density to describe it.

Type	Charge Density	Formula	SI Unit	Example
Linear (λ)	Charge per unit length	$\lambda = \frac{dq}{dl}$	$C m^{-1}$	Charged line
Surface (σ)	Charge per unit area	$\sigma = \frac{dq}{dA}$	$C m^{-2}$	Charged sheet
Volume (ρ)	Charge per unit volume	$\rho = \frac{dq}{dV}$	$C m^{-3}$	Charged body

1.13 Electric Field due to Continuous Distributions

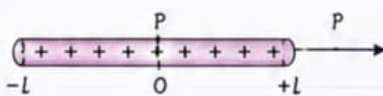
- (i) Due to a Charged Ring (radius R, charge Q)



$$E = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + R^2)^{3/2}}$$

(along the axis)

- (ii) Due to a Uniformly Charged Rod (length 2L, charge λ)



$$E = \frac{\lambda}{4\pi\epsilon_0 x} \left[\frac{x+L}{|x+L|} - \frac{x-L}{|x-L|} \right]$$

(along the axis)

- (iii) Due to a Uniformly Charged Disc (radius R, charge σ)



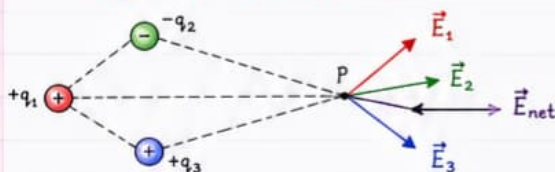
$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

(along the axis)

1.14 Superposition Principle

The electric field at any point due to a group of charges is the vector sum of electric fields due to individual charges.

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$



Quick Revision



- * Charge is the source, field is the effect.
- * Electric field is a vector quantity.
- * Field lines start from + and end on -.
- * Closer the lines, stronger the field.
- * Electric flux measures the number of field lines.
- * Gauss's law connects flux with enclosed charge.
- * Superposition principle is always valid.



1.15 Important Formulae at a Glance

Concept	Formula
Coulomb's Law	$F = \frac{1}{4\pi\epsilon_0} q_1 q_2 r^{-2}$
Electric Field (point charge)	$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
Field (infinite line)	$E = 2\pi\epsilon_0 \lambda r^{-2}$
Field (infinite sheet)	$E = \sigma / 2\epsilon_0$
Field (solid sphere, outside)	$E = 4\pi\epsilon_0 Q r^{-2}$
Field (solid sphere, inside)	$E = 4\pi\epsilon_0 Q r / R^3$
Dipole moment	$p = q \cdot 2a$
Dipole field (axial line)	$E = 4\pi\epsilon_0 \frac{2p}{r^3}$
Dipole field (equatorial line)	$E = 4\pi\epsilon_0 \frac{p}{r^3}$
Electric Flux	$\Phi_E = EA \cos \theta$ (uniform field)
Gauss's Law	$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$

Remember

→ Understand the concept clearly, practice numericals regularly and visualize field lines.

